

Abstract

The Martini force field, a set of interaction parameters for molecules, is widely used to study interactions between biological molecules. Its widespread use is due to it being a well-parameterized biological coarse-grained force field. Yet, many inorganic materials cannot be investigated in Martini because they have no direct correspondence to a Martini bead type. One such material is gold, making it infeasible to investigate gold-protein interactions, ligand-coated gold nanoparticle aggregation, and small molecule binding to gold surfaces in Martini. To bridge this gap, we present Martini Gold: a gold model and strategy for parameterizing metals in the Martini force field. We use atomistic simulations as a foundation for the Martini gold model. Our goal was to maintain gold cluster size, stability, and interaction energy between the coarse-grained and atomistic systems. To achieve this goal we determined the number of gold atoms that corresponds to a Martini S-bead, matched atomistic and coarse-grained gold-gold dimerization free energies in oil and water, and matched atomistic and coarse-grained small molecule adsorption free energies to gold surfaces. We validate the model using coarse-grained simulations of gold nanoparticle aggregation in lipid membranes and compare it to experimental cryo-TEM data. Based on the gold parameterization scheme, we designed a workflow to parameterize new metals in Martini.

Background

- Gold nanoparticles are used in biosensing, biolabelling, and controlled drug delivery
- Computational investigations of molecular interactions often use coarse-grained molecular dynamics, a low-resolution simulation method for investigating molecule interactions by mapping sets of atoms to beads.
- In Martini, a widely used biological coarse-grained force field (a set of molecular interaction parameters), there are no gold parameters.

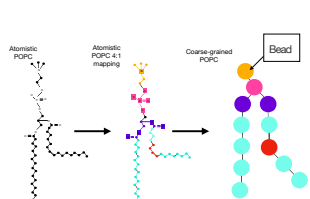


Fig 1. **Martini coarse-grained mapping scheme.** Martini molecules are on average mapped 4:1 from atomistic to a coarse representation of a molecule. Each coarse unit is called a bead and bead interactions are described by the composing atoms.

Model Goals

Develop a Martini model for gold that accurately captures gold-gold and gold biomolecule interactions

- What is the best mapping between a gold cluster and a Martini Bead?
- Can we reproduce the dimerization free energies of atomistic gold clusters?
- Can we reproduce the interaction free energies between gold surfaces and small molecules in aqueous environments?

Approach

- Match size of gold clusters** to a Martini bead
- Overlay Martini bead with an adjustable virtual site
- Match gold-gold dimerization free energies** between atomistic system and coarse-grained
- Match adsorption free energies for amino acids** to a gold slab and compare them between atomistic and coarse-grained systems
- Match adsorption free energies for lipids** to a gold slab and compare them between atomistic and coarse-grained systems

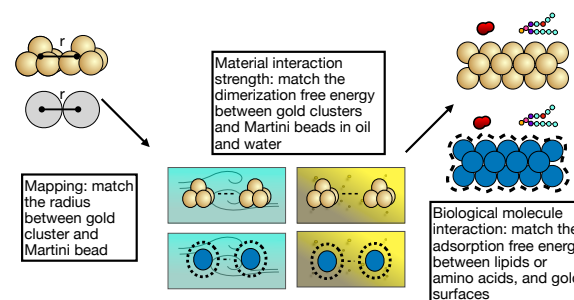


Fig 2. Martini coarse-grained gold model road map

Methods

- Simulation software: Gromacs 2024²
- Force Field: CHARMM³ with Interface⁴, Martini 3⁵
- System Building Tools: Charmm-gui^{7,8}, COBY⁹, Insane⁶
- Free energy profiles were obtained with COLVARS¹⁰
- All atomistic systems were equilibrated for 20 ns and production was 200 ns
- All coarse-grained systems were equilibrated for 200 ns and production was 2 μ s

6 atom gold cluster size matches the size of a Martini S bead

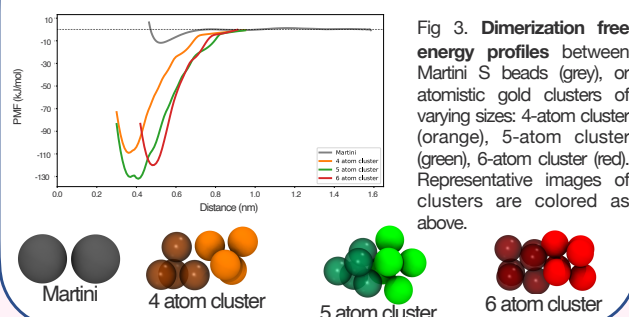


Fig 3. **Dimerization free energy profiles** between Martini S beads (grey), or atomistic gold clusters of varying sizes: 4-atom cluster (orange), 5-atom cluster (green), 6-atom cluster (red). Representative images of clusters are colored as above.

Tuning the Martini virtual site to match atomistic cluster dimerization free energy

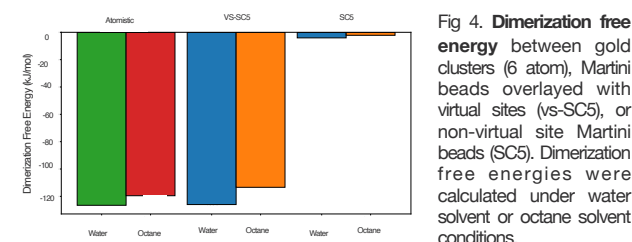


Fig 4. **Dimerization free energy** between gold clusters (6 atom), Martini beads overlayed with virtual sites (vs-SC5), or non-virtual site Martini beads (SC5). Dimerization free energies were calculated under water solvent or octane solvent conditions.

Small molecule adsorption free energies used to tune the gold model's behavior in biological environments

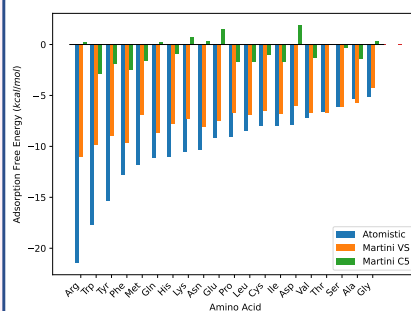


Fig 5. **Adsorption free energy of the 20 amino acids to a slab of:** atomistic gold (blue), Martini bead overlayed with virtual site tuned to newly specified potentials for Martini gold (orange), or Martini C5 beads (green). Molecular image from a simulation of an amino acid (red) and a gold slab (gold) in a water box (cyan).

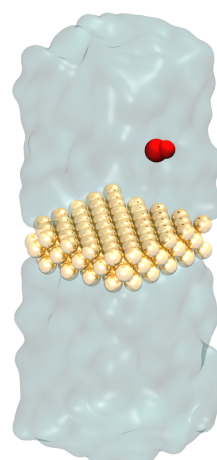


Fig 6. **Representative image of gold nanoparticle model** built with the adjusted Martini virtual site gold potentials.

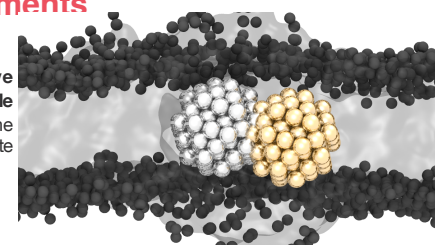


Fig 7. **Adsorption free energy of lipid pieces to a slab of:** atomistic gold (blue), Martini bead overlayed with virtual site tuned to newly specified potentials for Martini gold (orange)

Summary

- Here we have developed the first Martini gold model that captures gold-gold and gold-biomolecule interactions.
- We match several targets to appropriately model gold cluster size, dimerization propensity, and small molecule adsorption.
- We tested the model by building a gold nanoparticle system and observed strong gold nanoparticle dimerization, which is expected from qualitative observation (Fig. 6, 8).

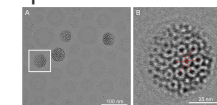


Fig 8. Cryo-TEM images of GNPs aggregating in polymersomes

References

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